

sensor_counted

How sensor_counted was designed and verified, with every number read from its recorded gate results.

What it is

A ring oscillator into a counter (docs/design.md section 3's /msde corpus list). The analog block free-runs a ring oscillator whose frequency the digital side counts over a fixed gate window; a digital enable powers the oscillator down between measurements.

This rung exercises the split gate (docs/design.md 1.5 msde table): the splitter's interface.yaml names every crossing signal, and the digital and analog sides each carry their own copy of that list in digital_spec.yaml / analog_spec.yaml (design.md section 5: "two specs carrying the same entries, and split checks they agree").

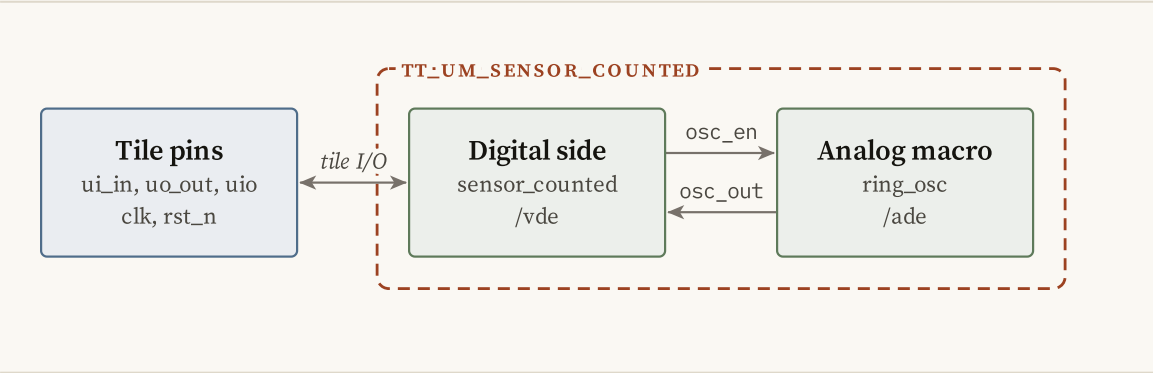
The two nested /vde and /ade workspaces this block would drive at P2 (docs/design.md 1.4) are not built at M10 - they land once M4 and M9 merge. digital_spec.yaml / analog_spec.yaml here are the two sides' declared interface only, standing in for a real nested block's own spec.yaml interface: section until then.

SIGNALS CROSSING THE SPLIT, FROM INTERFACE.YAML

SIGNAL	WAY	LEVEL	DOMAIN	LOAD
osc_out	a2d	cmos_3v3	clk_free	digital 2-flop synchroniser input, CMOS gate load <= 50 fF
osc_en	d2a	cmos_3v3	clk_sys	analog enable gate input, static level, no timing constraint

Architecture, and why

FIGURE 1 *block diagram*



Drawn from `interface.yaml`.

The decisions recorded while the block was designed, each with its reason:

- **cosim bench** patches `cocotbext-ams NgspiceInterface._on_get_sync_data` to snap sync points under 1 ps cocotbext-ams sync clamp float rounding handed ngspice a 1e-23 s step (Timestep too small at 100 ns) on the transistor-level `ring_osc`; the guard changes no device, stimulus or bound; an engine-level fix in the bridge is a follow-up (P3)

How it was verified

6/6	39	0
gates with a fresh recorded pass	counts latched in cosim, 38.35 expected	top-level DRC violations

TABLE *the msde gates, as recorded*

GATE	CHECKS	RESULT	RECORDED NUMBERS
cosim	mixed-signal cosim	pass	count 39 vs 38.35 expected; f_{osc} 1.498e6 Hz 2026-09-25 11:10
precheck	Tiny Tapeout precheck	pass	12 checks run 2026-09-25 10:58
split	interface agrees	pass	2 crossing signals agree 2026-09-25 03:24
top_drc	top-level DRC	pass	magic 0, klayout 0 violations 2026-09-25 10:52
top_harden	top-level harden	pass	201 s 2026-09-25 10:49
top_lvs	top-level LVS	pass	match; 28/28 macro FETs 2026-09-25 10:52

For top_harden the full result of the run on record was not kept, so the row gives only the numbers state.json recorded with it.

The digital side, sensor_counted

Released. reports/checks.json verifies against the current files and gate records (attestation 432b9e1aff1e). Human sign-off H2: approved on 2026-09-25. 0 waivers recorded.

TABLE *the digital side's vde gates*

GATE	CHECKS	RESULT	RECORDED NUMBERS
cover	code coverage	pass	line 100.0% (72/72); toggle 94.9% (112/118) 2026-09-25 09:29
drc	magic + klayout DRC	pass	magic 0, klayout 0 violations 2026-09-25 09:00
formal	SymbiYosys proof	pass	3 properties proven; depth 24; smt PASS; pdr PASS; 1 covers reached 2026-09-25 03:56
glsim	gate-level sim	pass	0 findings, 0 failing 2026-09-25 09:24
harden	LibreLane to GDS	pass	182 s 2026-09-25 08:49
holdout	held-out tests	pass	7/7 tests pass 2026-09-25 04:01
lint	Verilator lint	pass	0 findings, 0 failing 2026-09-25 03:56
lvs	netgen LVS	pass	0 findings, 0 failing 2026-09-25 09:01
mutate	mutation score	pass	17/17 killed; kill rate 100.0%; 6 ruled equivalent 2026-09-25 09:42
precheck	Tiny Tapeout precheck	pass	12 checks run 2026-09-25 09:26
sim	cocotb simulation	pass	12 tests pass 2026-09-25 09:28
spec_lint	spec is checkable	pass	0 findings, 0 failing 2026-09-25 09:33
synth	Yosys synthesis	pass	area 5515 μm^2 2026-09-25 08:19

timing	OpenSTA signoff	pass	worst setup slack 69.26 ns, hold 0.38 ns over 9 corners 2026-09-25 08:57
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For cover, drc, glsim, harden, lvs, mutate, precheck, sim, spec_lint, synth, timing the full result of the run on record was not kept, so the row gives only the numbers state.json recorded with it.

Performance, power and area

Synthesised cell area is 5515 μm^2 on the GF180 standard cells (synth gate). At a 100 ns clock the worst setup slack is 69.26 ns and the worst hold slack 0.38 ns, over 9 signoff corners (timing gate). No gate in this flow measures power, so this document gives no power figure.

TIMING SLACK PER CORNER, NS

CORNER	SETUP	HOLD
max_ff_n40C_3v60	76.83	0.39
max_ss_125C_3v00	69.26	2.06
max_tt_025C_3v30	74.58	0.89
min_ff_n40C_3v60	76.92	0.38
min_ss_125C_3v00	69.54	2.04
min_tt_025C_3v30	74.73	0.88
nom_ff_n40C_3v60	76.88	0.39
nom_ss_125C_3v00	69.41	2.05
nom_tt_025C_3v30	74.66	0.88

The analog side, ring_osc

Released. reports/checks.json verifies against the current files and gate records (attestation c8d6c27c6a0c). Human sign-off H2: approved on 2026-09-25. 0 waivers recorded.

TABLE *the analog side's ade gates*

GATE	CHECKS	RESULT	RECORDED NUMBERS
bench_strength	bench mutants	pass	68/78 killed; 6 ruled equivalent; 10 below spread; 16 survived 2026-09-25 10:32
drc	magic + klayout DRC	pass	0 findings, 0 failing 2026-09-25 10:24
lvs	netgen LVS	pass	0 findings, 0 failing 2026-09-25 10:24
mc	Monte Carlo	n/a	not applicable: spec.yaml asks for no Monte Carlo (no 'mc.enabled: true'), re-read at this release
netlist_lint	ngspice dry run	pass	0 findings, 0 failing 2026-09-25 10:23
pex_sim	post-layout sim	pass	0 findings, 0 failing 2026-09-25 10:25
sim_pvt	PVT corners	pass	5 corners: tt, ss, ff, sf, fs 2026-09-25 10:26
sim_tt	typical corner	pass	1 corners: tt 2026-09-25 10:25
spec_lint	spec is checkable	pass	0 findings, 0 failing 2026-09-25 10:23

For bench_strength, drc, lvs, netlist_lint, pex_sim, sim_pvt, sim_tt, spec_lint the full result of the run on record was not kept, so the row gives only the numbers state.json recorded with it.

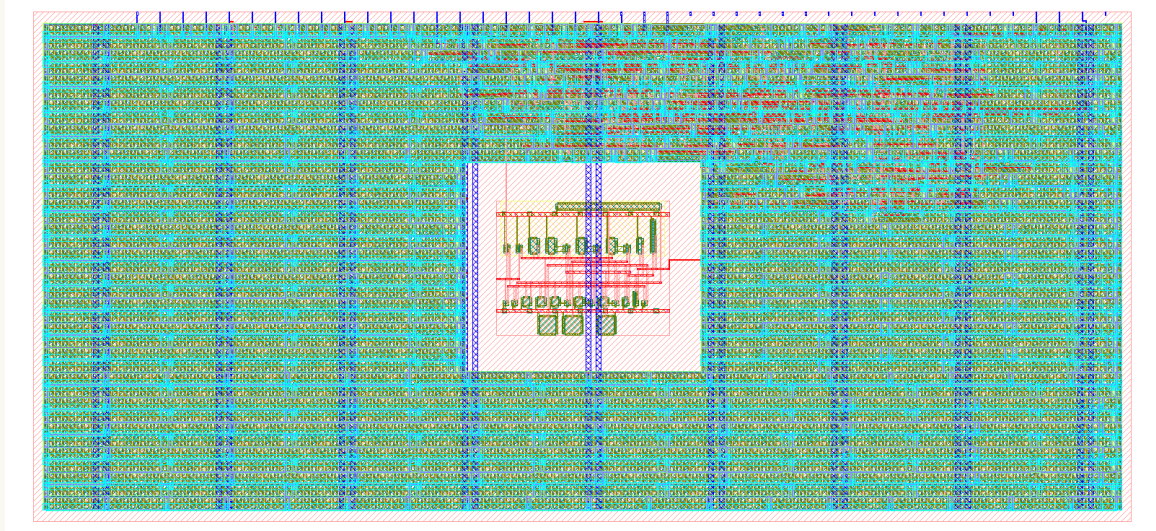
The sim_pvt run on record kept no per-measure values (its full result was not saved with it), so the table gives the bounds it passed against and no measured numbers.

ANALOG MEASURES, BOUNDS FROM TB/*.BOUNDS.JSON, THE BOUNDS THE GATE SCORES

MEASURE	BOUND	MEASURED, MIN TO MAX	CORNERS
f_osc_hz	1.348e6 to 1.649e6	not kept	tt
f_osc_hz_pvt	5.000e5 to 4.500e6	not kept	all
t_startup_s	$\leq 5.000\text{e-}6$	not kept	all
duty_pct	52.55 to 56.02	not kept	tt
t_rise_s	$\leq 5.000\text{e-}9$	not kept	all
t_fall_s	$\leq 5.000\text{e-}9$	not kept	all
v_out_low_v	≤ 0.3	not kept	all
v_out_high_v	≥ 2.7	not kept	all
i_vdd_en_a	2.119e-5 to 2.326e-5	not kept	tt
i_vdd_dis_a	5.220e-11 to 5.834e-11	not kept	tt
i_vdd_dis_a_pvt	$\leq 1.000\text{e-}6$	not kept	all
t_powerdown_s	1.423e-9 to 1.867e-9	not kept	tt
t_powerdown_s_pvt	$\leq 1.000\text{e-}6$	not kept	all

Layout

FIGURE 2 *layout*



top/harden/runs/run/final/gds/tt_um_sensor_counted.gds, rendered with layout.

Release record

Released. reports/checks.json verifies against the current files and gate records (attestation 6f9240016508). Human sign-off H2: approved on 2026-09-25. 0 waivers recorded.